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Colab Script:
https://colab.research.google.com/drive/1apcjIqmdiSen5BnDShe5h0jMeMaNRdSz?usp=sharing

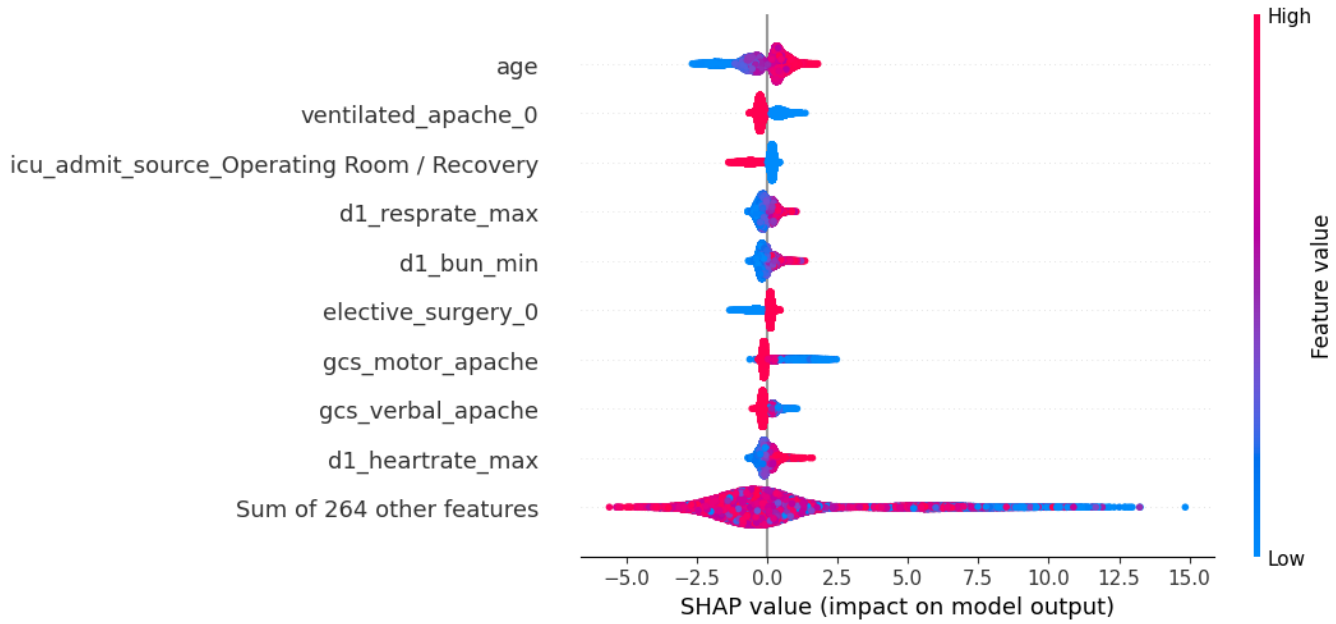
Part 1. Mortality Prediction in the ICU

(f) Which model has the best performance? Briefly explain your answer.

After evaluating various models (LR, RF, XGB), the gradient-boosted random forest classifier (XGB) demonstrated the best performance on the test set. This conclusion is based on key metrics such as the F1 score and the AUC score, which are particularly important given the imbalance in our dataset. These metrics provide a robust assessment of the model's ability to handle imbalanced classes, with the F1 score balancing precision and recall, while the AUC score measures the model's capacity to distinguish between positive and negative classes across different thresholds. However, it's worth noting that the XGB model exhibited near-perfect performance on the training set, indicating potential overfitting. This suggests that while the model generalizes well to the test set, it may be overfitting to the training data. To address this, a more thorough grid search to fine-tune the hyperparameters could be beneficial in improving model generalization and reducing the risk of overfitting.

(g) To get a high-level overview of which features contribute the most to our models' predictions, use the shap library to

compute and plot the Shapley values of the xgboost model's features on a beeswarm plot. Which features contribute the most to the model's predictions? Do they seem like reasonable features that the model can rely on, or is the model basing its predictions on spurious correlations?



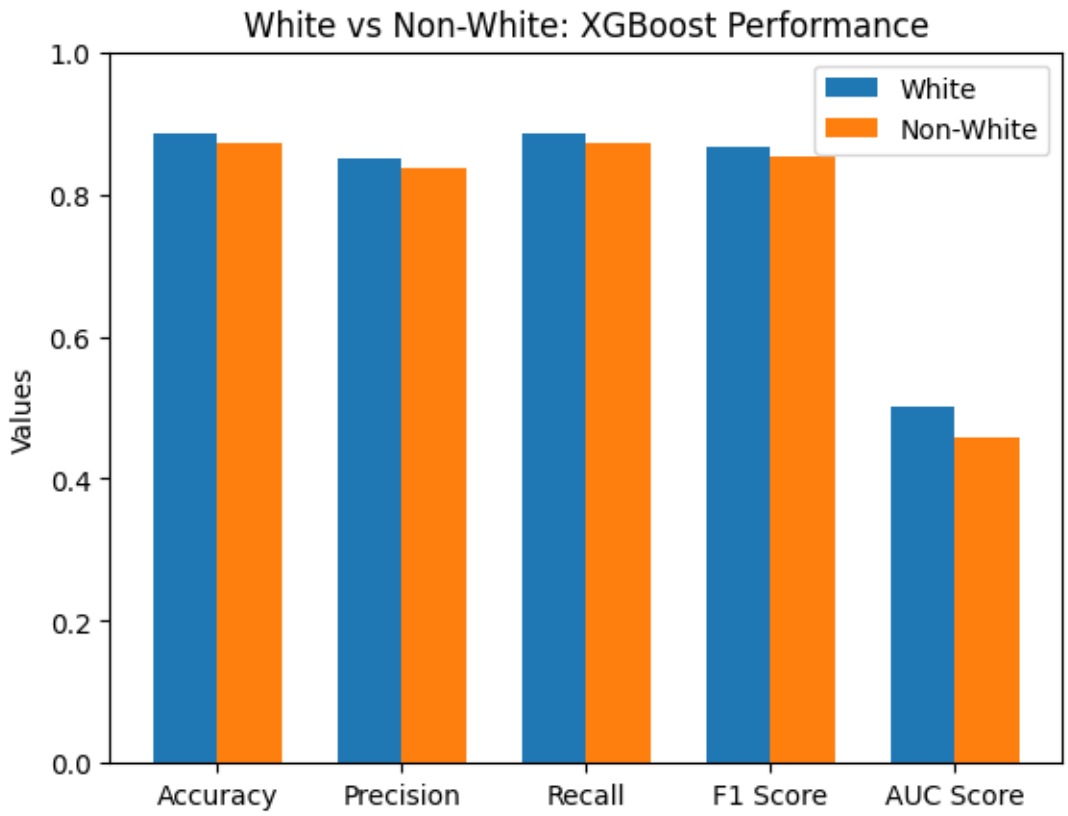
The key features identified by the model, and their contributions based on SHAP values, are as follows:

- Age: Age is a reasonable feature, with older age correlating with a higher probability of death. This aligns with established clinical knowledge that mortality risk increases with age in critically ill patients.
- Ventilation status: The model suggests that being ventilated is associated with a lower mortality risk. While this may initially seem counterintuitive, it could imply that non-invasive ventilation (as opposed to intubation) is being captured by this feature. Generally,

intubated patients tend to have worse outcomes, so this distinction may explain the association.

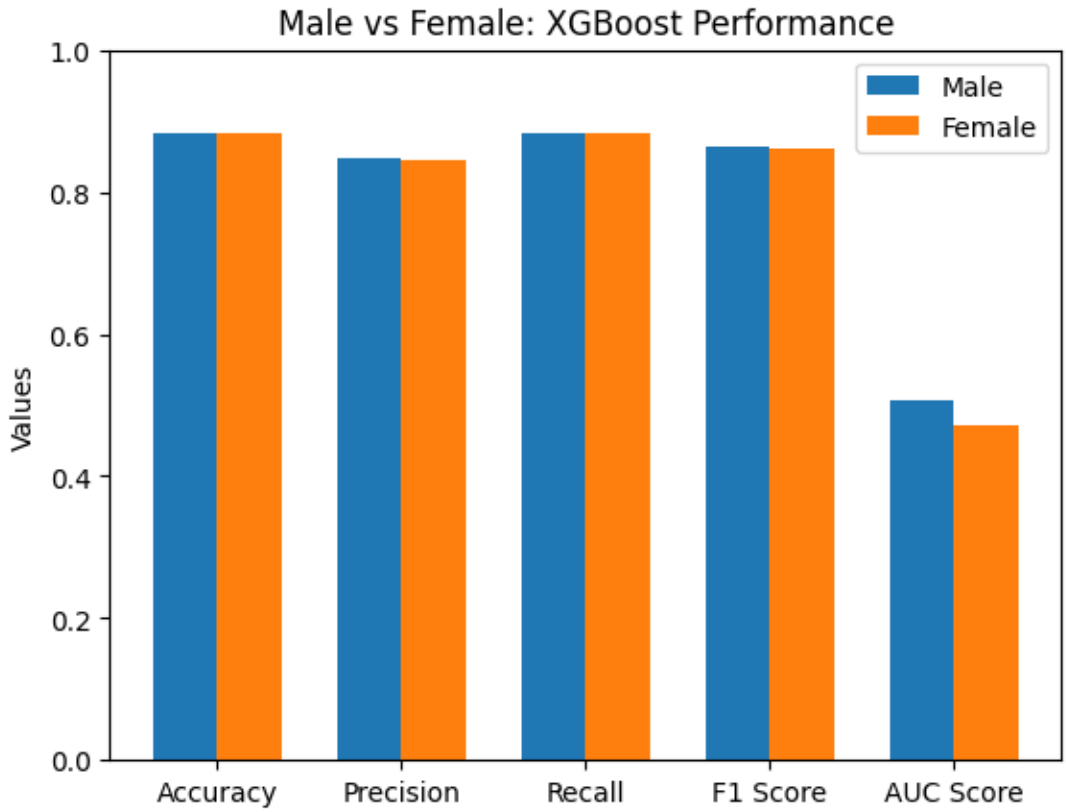
- Respiratory rate and heart rate: Both high respiratory and heart rates are predictors of respiratory failure, which makes their association with mortality quite reasonable. These features are clinically relevant in the ICU setting, as they often indicate the severity of illness.
- Blood Urea Nitrogen (BUN): Elevated BUN levels are strongly associated with mortality in ICU patients, particularly in those experiencing acute exacerbations of chronic obstructive pulmonary disease (COPD). (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9699438/)
- Glasgow Coma Scale (GCS): Both the motor and verbal components of the GCS show a strong inverse relationship with mortality. Higher GCS scores, indicating better consciousness levels, are less associated with mortality, which is consistent with the clinical understanding that lower GCS scores are linked to loss of consciousness and coma.
- Elective surgery status: Patients who did not undergo elective surgery are at higher risk of death. This makes sense as elective surgeries are typically not performed on unstable patients, meaning this feature likely acts as a proxy for overall patient health and stability.

(j) How well does the model perform on the two cohorts in each split?



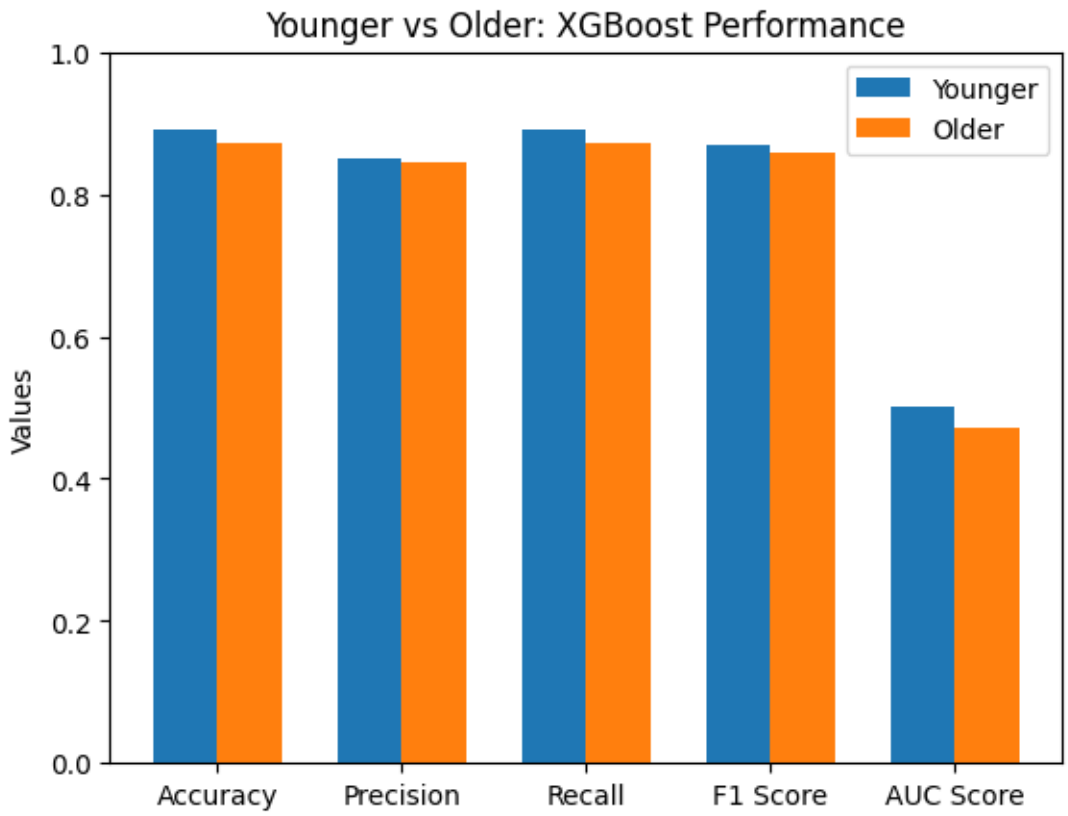
When evaluating the model's performance across the two cohorts split by ethnicity, it is apparent that the model performs significantly better for white patients compared to the group encompassing all other ethnicities. This disparity is evident across all evaluation metrics utilized in our analysis. Such findings raise important considerations regarding the model's generalizability and fairness, suggesting that it may be disproportionately benefiting one demographic group. This underscores the necessity for further investigation

into potential biases in the training data or feature representation.



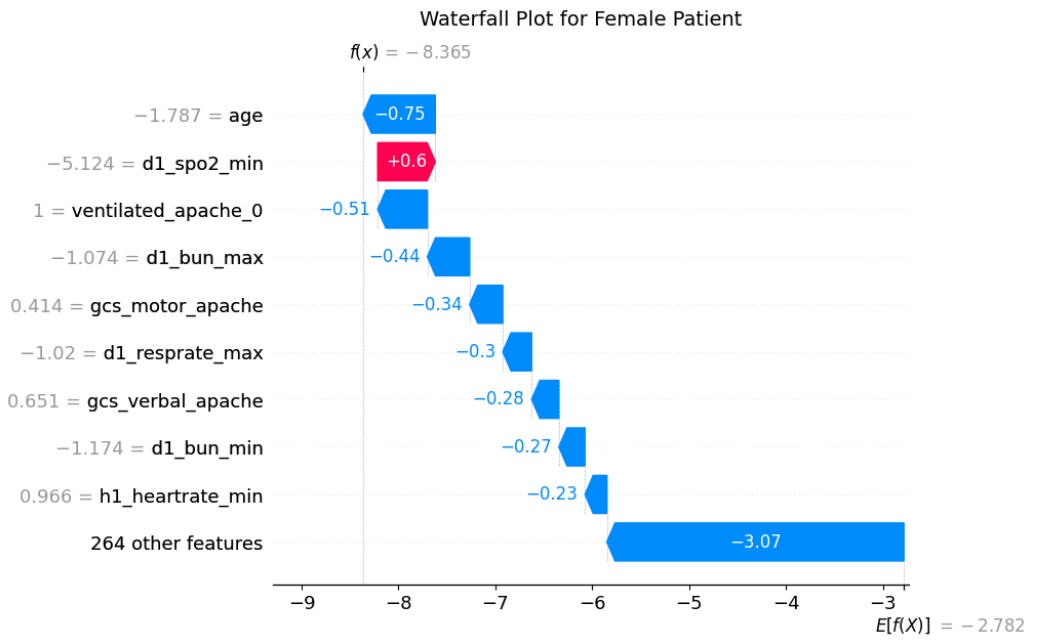
The performance of the model on male and female patients is generally comparable across all evaluation metrics. However, a notable exception is observed in the AUC score, where the XGBoost model demonstrates a greater decline in performance for female patients compared to their male counterparts.

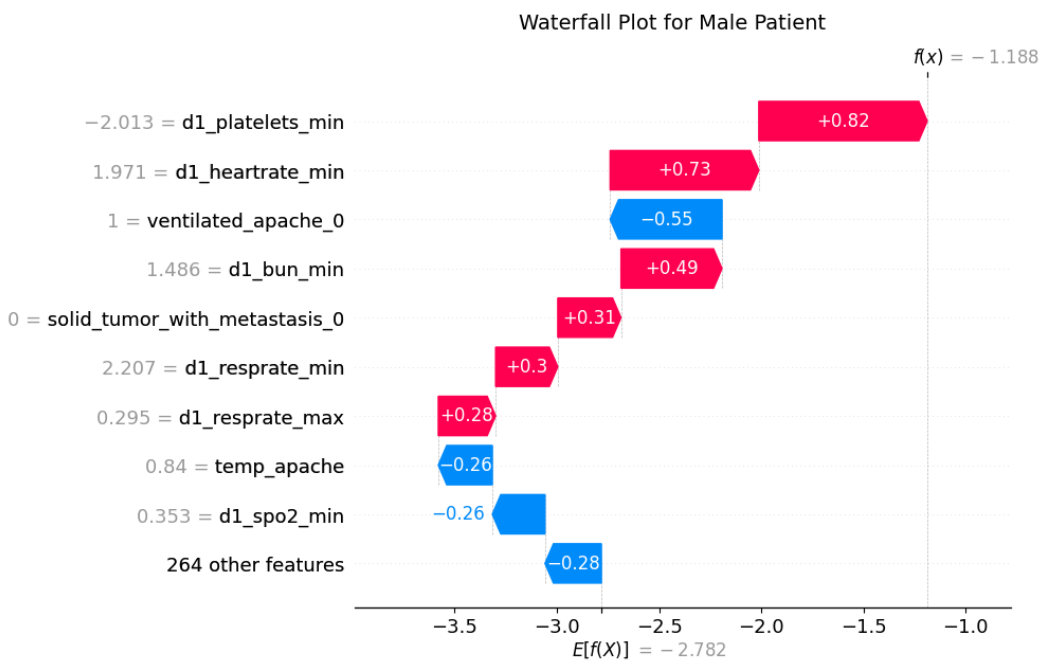
This discrepancy in AUC scores highlights a potential area of concern regarding the model's robustness and ability to generalize effectively across genders. It suggests that while the model performs adequately overall, there may be underlying factors contributing to its diminished performance for females.



Lastly, the performance of the model is slightly worse across all evaluation metrics for older patients (>65 years) compared to their younger counterparts (<65 years). This trend raises concerns about the model's effectiveness in predicting outcomes for the older demographic, who are often at higher risk of mortality. Moreover, it is particularly alarming that the AUC scores are hovering around or even dipping below the 0.5 threshold across various splits. Such scores indicate a model that is no better than random chance, necessitating further investigation.

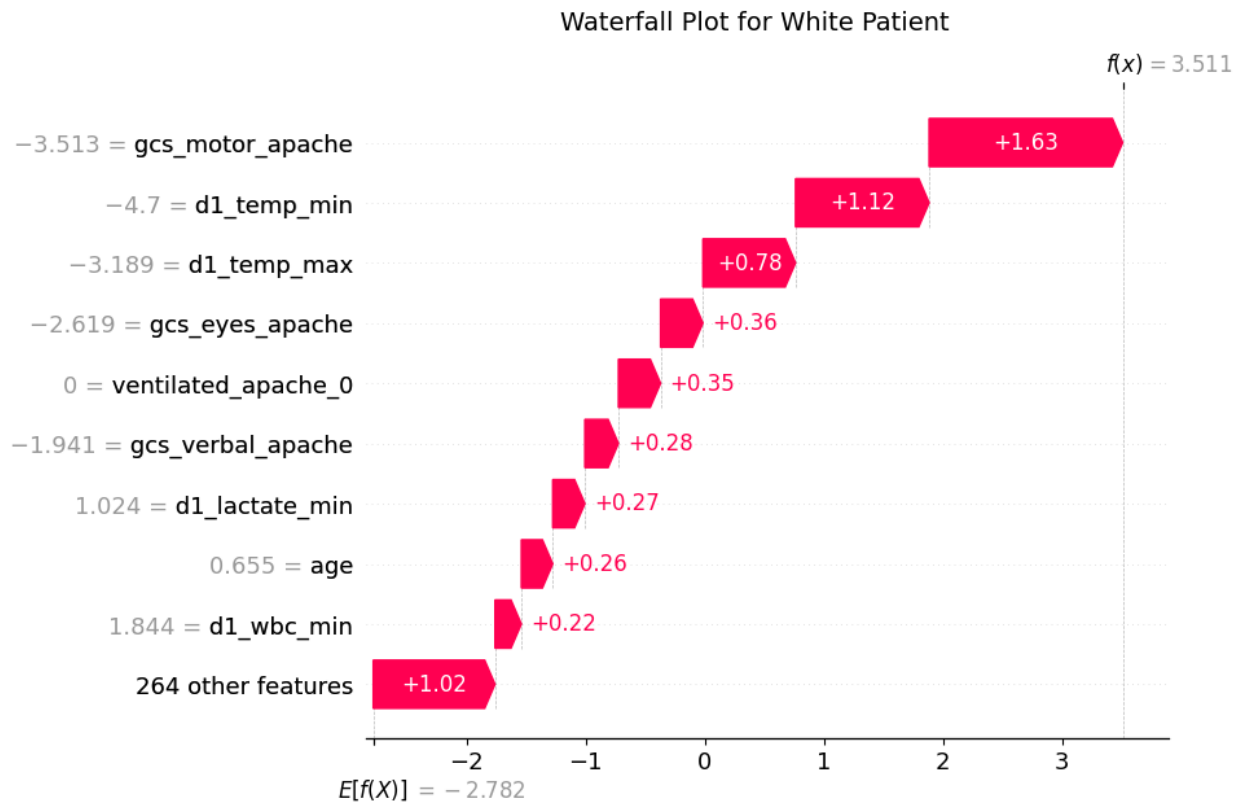
(l) <u>Do you notice any discrepancies in the features used by the model to make predictions for the two cohorts in each split? If yes, briefly describe those discrepancies. If not, briefly explain why you think such discrepancies were not observed.</u>

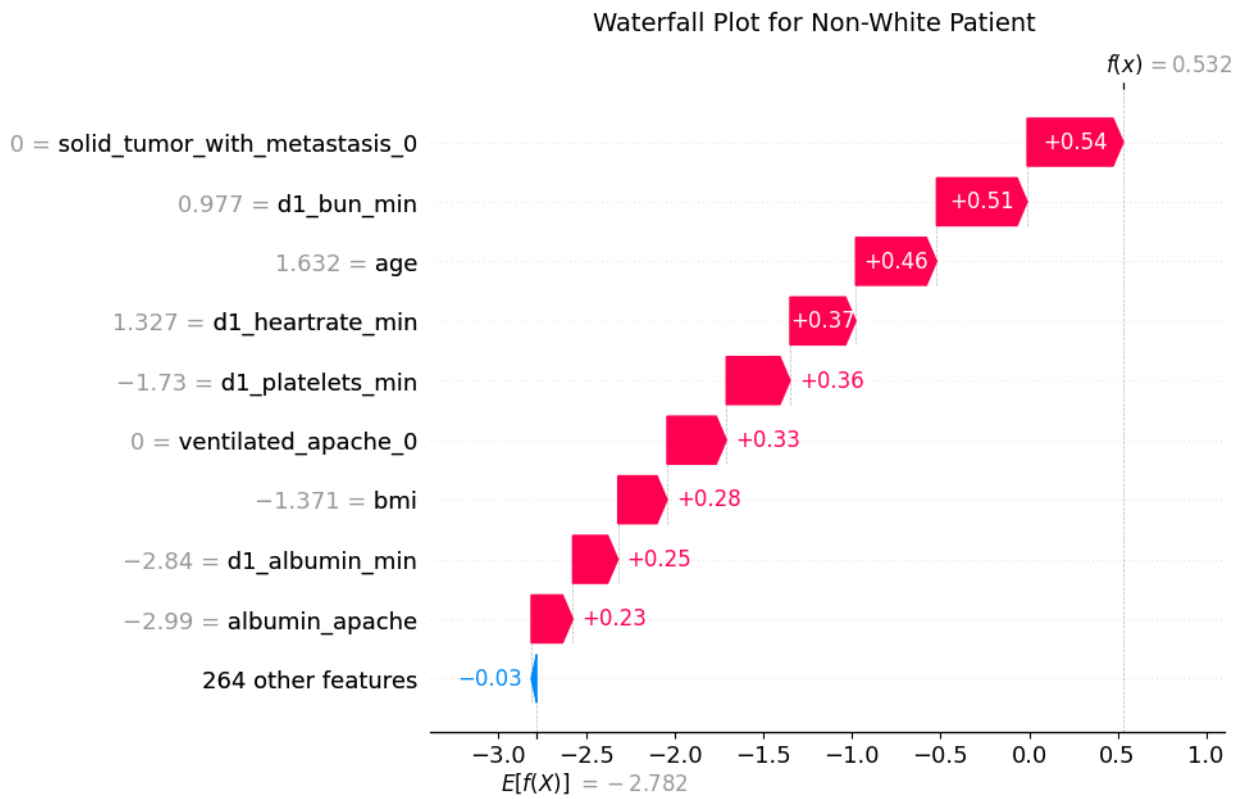




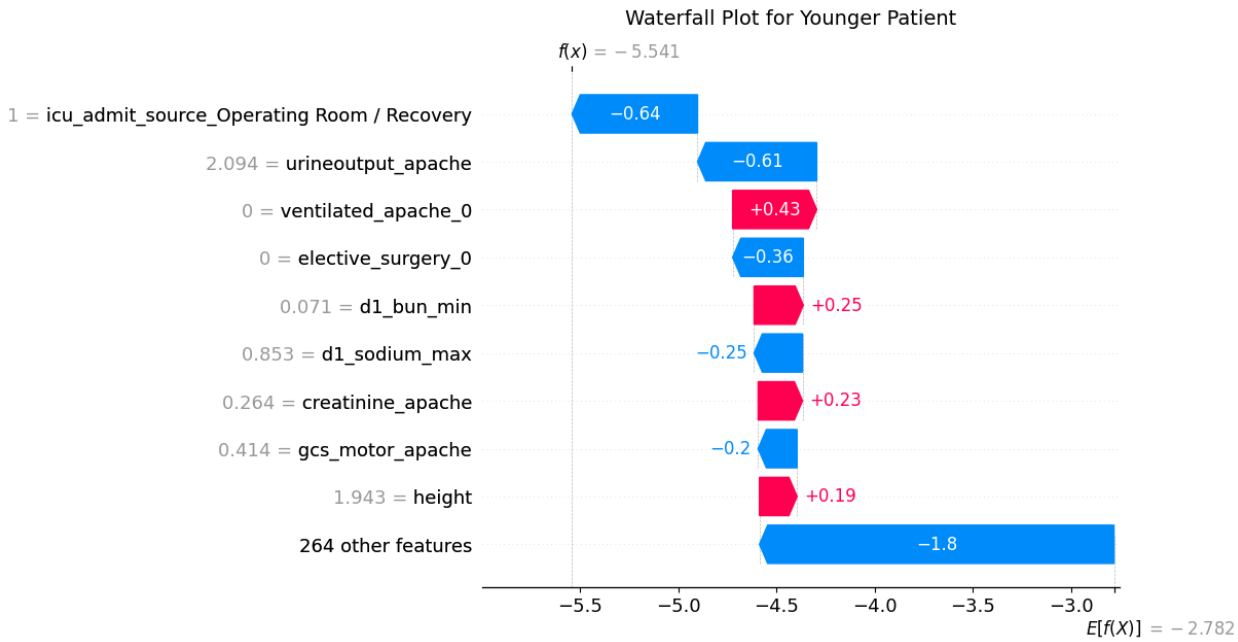
While it is challenging to draw definitive conclusions from a limited number of examples, some discrepancies in feature contributions can be observed between the male and female cohorts. Specifically, the model's prediction for the male patient appears to rely on only a few guiding features, with the total contribution of the 264 unlisted features summing to merely -0.28. In

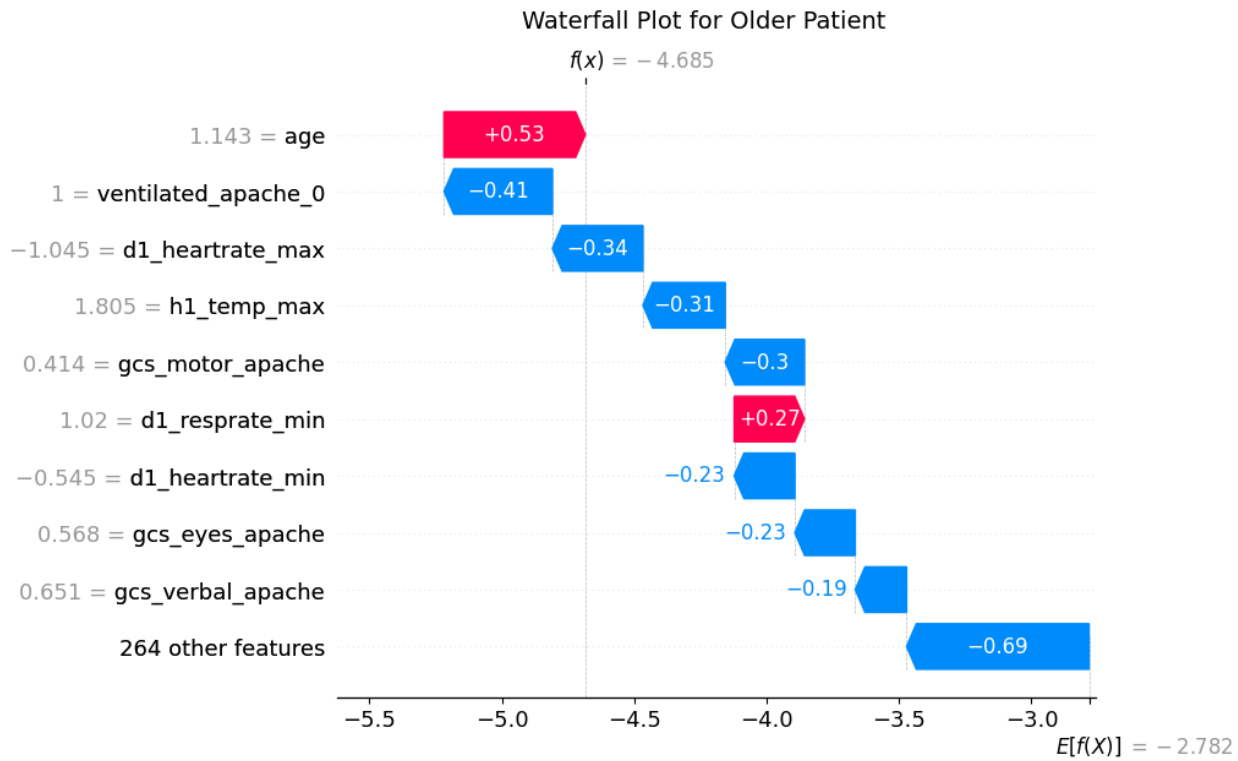
contrast, the prediction for the female patient is influenced by a more substantial contribution of -3.07 from the unlisted features. This disparity suggests that the model may have lower interpretability for female patients, as a broader array of features appears to drive predictions in this cohort.





The waterfall plot for the white patient presents a clearly interpretable and clinically coherent scenario. The prediction of mortality is driven by several key factors, including low scores in all components of the Glasgow Coma Scale (GCS), advanced age, elevated lactate levels indicative of sepsis, and low body temperature. These factors collectively align with established clinical knowledge regarding severe illness and mortality risk. In contrast, the interpretation for the non-white patient is less straightforward. The primary contributor to the predicted mortality is the absence of metastasis, which appears contradictory in the context of typical clinical reasoning. This unexpected result raises concerns about the model's reliance on potentially spurious correlations or biases in feature importance across different cohorts.





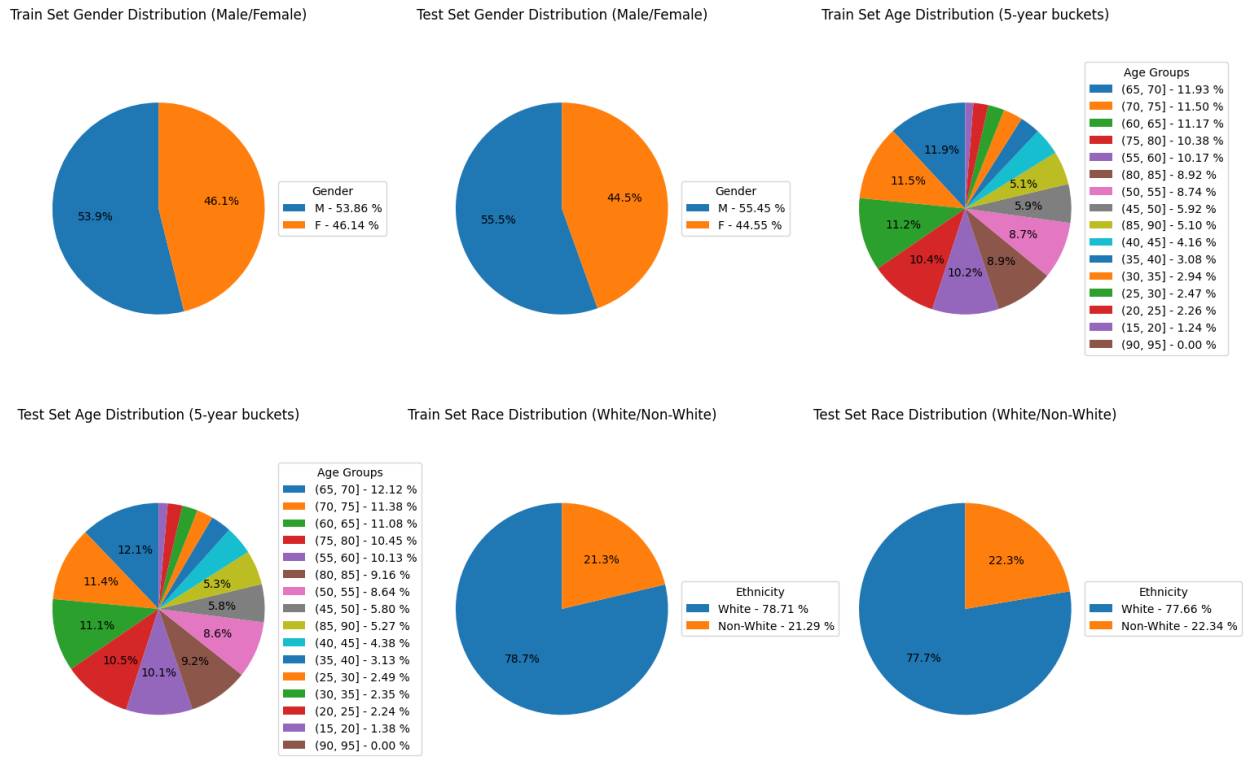
When comparing the explanations for the single examples of older and younger patients, the age feature significantly influences the prediction for the older patient, often driving the mortality risk in the opposite direction. Even though this older patient presents with high GCS scores and appears stable, their advanced age can still lead the model to increase the probability of

mortality. In contrast, the interpretability of the model's prediction for the younger patient is quite challenging based on this single example. Multiple features are driving the model in conflicting directions, leading to inconsistencies and potentially spurious associations. For instance, in this case, the model suggests that greater height is associated with an increased mortality risk, which is a spurious association.

Part 2. Delving into Disparities

(a) Plot, each on a different pie chart, the distribution of the patients' genders, ages (bucketed into 5 year intervals), and races for each of the train and test splits (6 pie charts in total). Do you notice any imbalances in the data?

There are some imbalances in the data, with a higher proportion of males in the training set that is even higher in the test set. The imbalance between Caucasian ethnicity and other ethinicities is also prominent, with around 80% of the patients being white.



(b) We would like to check if varying the degree of imbalance in the GOSSIS dataset has an effect on the performance of a model trained on the imbalanced dataset. Follow the instructions in the notebook to reduce the number of training datapoints corresponding to female patients by 20%, 40%, 60%, and 80% respectively. For each reduction in the number of datapoints, train an xgboost model on the modified dataset and compute the same performance metrics introduced in part 1 (accuracy, precision, recall, F1 score, and AUC score) on both the modified training set and the original test set. Additionally, compute the same performance metrics on each of the male and female cohorts. Report your results in a table.

Percentage Drop	Cohort	Accuracy Train	Precision Train	Recall Train	F1 Train	AUC Train	Accuracy Test	Precision Test	Recall Test	F1 Test	AUC Test
0.2	Overall	0.999	0.999	0.999	0.999	1.0	0.928	0.917	0.929	0.917	0.886
0.2	Male	0.999	0.999	0.999	0.999	1.0	0.929	0.918	0.929	0.918	0.889
0.2	Female	0.999	0.999	0.999	0.999	1.0	0.927	0.916	0.928	0.916	0.882
0.4	Overall	0.999	0.999	0.999	0.999	1.0	0.928	0.916	0.928	0.916	0.881
0.4	Male	0.999	0.999	0.999	0.999	1.0	0.929	0.917	0.929	0.917	0.883
0.4	Female	0.999	0.999	0.999	0.999	1.0	0.928	0.916	0.928	0.915	0.880
0.6	Overall	0.999	0.999	0.999	0.999	1.0	0.928	0.916	0.928	0.915	0.881
0.6	Male	0.999	0.999	0.999	0.999	1.0	0.930	0.918	0.930	0.917	0.886
0.6	Female	0.999	0.999	0.999	0.999	1.0	0.926	0.914	0.926	0.912	0.875
0.8	Overall	0.999	0.999	0.999	0.999	1.0	0.929	0.918	0.929	0.917	0.879
0.8	Male	0.999	0.999	0.999	0.999	1.0	0.931	0.920	0.931	0.920	0.885
0.8	Female	0.999	0.999	0.999	0.999	1.0	0.927	0.914	0.927	0.913	0.872

(c) How does reducing the number of female patient datapoints affect the performance of the model? Why do you think that is?

Reducing the number of female patient data points in the training set adversely affects the performance of the model on the test set, particularly noticeable in the AUC metric. While the overall decline in performance may be minimal, it becomes more pronounced when examining the two subgroups. As expected, the performance for the female cohort progressively decreases with the removal of data points. Interestingly, the male cohort also experiences a slight reduction in performance, which is notable. This indicates that the model may be leveraging information from both cohorts during training, and a decrease in one cohort's representation can impact overall performance. The primary issue at play is that the XGBoost model is overfitting to the training dataset. This overfitting, which is already evident before the dataset is split, persists after the splits.

(d) Do you expect to see the same or different results when varying the degree of missingness of some populations in

other datasets or cohort splits? Briefly explain why you think so.

I would expect to see similar results when varying the degree of missingness in certain populations, particularly concerning ethnicity. In our dataset, where approximately 80% of patients are white, the representation of minority ethnic groups is already sparse. Reducing the data points for these underrepresented groups would likely exacerbate performance disparities. Additionally, considering age groups, if there is a lack of representation for older patients, who are often at higher risk, the model may underperform when predicting outcomes for that demographic. Similar to our observations with the female cohort, a decline in the performance of the model can be expected when data points from either ethnic or age groups are reduced.

(e) Another form of data imbalance is more implicit, and happens due to some patients undergoing fewer tests even if the number of patients is the same in each cohort. How do you expect increasing the degree of missingness of test results in some training datapoints would affect the performance of your model? Briefly justify your answer.

Increasing the degree of missingness in test results, particularly when this missingness occurs not at random (MNAR), is likely to have a detrimental effect on the model's performance. In healthcare, the absence of specific test results can convey critical information about a patient's condition. For instance, if test results are less frequently recorded for patients from minority ethnic groups—perhaps due to biases in clinical practice or resource allocation—this could lead to significant information loss. Also, missingness could be related to patients being perceived as stable or less severe, particularly among underrepresented groups. If the model is trained without appropriately accounting for this MNAR scenario, it may misinterpret the

absence of results as unimportant, leading to skewed predictions and biased assessments of feature importance.

(f) Suggest one potential way of handling missingness of test results.

A critical first step in addressing the missingness of test results is to include the individuals involved in data collection during the early-stages of model development. Their proximity to where the data are collected and their awareness of the underlying reasons for missing data are essential for identifying potential biases and improving data quality. One effective approach to managing the missingness of test results is to incorporate missingness indicators into the model. This method provides additional context, allowing the model to distinguish between cases where a test was simply not conducted and cases where the absence of a test result may be not at random. In conclusion, simply ignoring missing values or relying on traditional imputation methods can exacerbate the problem, as they may not adequately capture the complexities and biases associated with the missing data. By addressing MNAR, we can enhance the robustness and fairness of machine learning models in healthcare.